

Bayesian Bike-Sharing Project — Module B (Sections 2.2 and 3.1)

0.1 Likelihood and Prior

Likelihood (Core Task 1, §3.1) Let y_i denote the daily rental count and let $x_{i1}, x_{i2} \in [0, 1]$ denote the min–max normalised temperature and humidity for observation $i = 1, \dots, n$ with $n = 360$. The empirical distribution of y is approximately symmetric with sample mean $\bar{y} = 4392$, sample standard deviation $s_y = 1936$, and range $[22, 8555]$. We adopt a homoscedastic Gaussian linear regression as the sampling model:

$$y_i \mid \boldsymbol{\beta}, \sigma \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_i, \sigma^2), \quad \mu_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}, \quad \boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^\top. \quad (1)$$

Two natural alternatives were considered and rejected. A Poisson likelihood imposes equidispersion ($\text{Var} = \mathbb{E}$); the empirical variance-to-mean ratio of y exceeds unity by an order of magnitude, so a Poisson specification would yield artificially narrow credible intervals. A Negative Binomial likelihood relaxes this constraint via an extra dispersion parameter but introduces a non-Gaussian link without a corresponding gain in fit, given the approximate symmetry of y . The Gaussian specification (1) is therefore preferred and the Negative Binomial is retained as a robustness-check candidate.

Prior specification (Core Task 1, §3.2) We assign mutually independent weakly informative priors,

$$\beta_j \sim \mathcal{N}(0, \tau_\beta^2), \quad j = 0, 1, 2, \quad \sigma \sim \text{Half-}\mathcal{N}(\tau_\sigma), \quad (2)$$

with hyperparameters $\tau_\beta = 10,000$ and $\tau_\sigma = 3,000$. Under (2) each β_j has prior mean, median, and mode equal to zero, prior standard deviation τ_β , and 95% prior credible interval $\approx (-19,600, 19,600)$. The Half-Normal prior on σ enforces the positivity constraint $\sigma > 0$ and yields the closed-form summaries $\mathbb{E}[\sigma] = \tau_\sigma \sqrt{2/\pi} \approx 2394$, $\text{SD}(\sigma) = \tau_\sigma \sqrt{1 - 2/\pi} \approx 1808$, prior median $\tau_\sigma \Phi^{-1}(0.75) \approx 2024$, and 95-th percentile ≈ 5880 (Table 1).

Table 1: Prior distributions and implied measures of location and dispersion.

Parameter	Distribution	Hyperparameter(s)	Prior mean	Prior SD
$\beta_0, \beta_1, \beta_2$	Normal	$\tau_\beta = 10,000$	0	10,000
σ	Half-Normal	$\tau_\sigma = 3,000$	≈ 2394	≈ 1808

Calibration of hyperparameters Both hyperparameters are calibrated on the empirical scale of the data. Since each x_{ij} is normalised to $[0, 1]$, a unit change in covariate j corresponds to its full empirical range, and plausible coefficient magnitudes are bounded above by $\max_i y_i - \min_i y_i \approx 8.5 \times 10^3$. Setting $\tau_\beta = 10,000$ places approximately 95% of prior mass on $|\beta_j| < 19,600$, enveloping the ordinary-least-squares (OLS) estimates $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2) \approx (2548, 6796, -2370)$ while excluding only values that would imply effects exceeding the empirical range of y . The choice $\tau_\sigma = 3,000$ aligns the prior median for σ (≈ 2024) with the empirical scale $s_y \approx 1936$, accommodates the OLS residual standard error $\hat{\sigma}_{\text{OLS}} = 1497$ in a region of substantial prior mass, and avoids the small-scale pathologies of the Inverse-Gamma. By design, τ_β and τ_σ are an order of magnitude larger than the posterior standard deviations reported in Section 0.2, so the likelihood dominates the posterior and Section can meaningfully assess sensitivity to the prior.

Prior predictive check To verify that (2) is reasonable on the natural scale of the response, we drew 500 samples from the joint prior predictive distribution of y (Figure 1). The resulting distribution is approximately symmetric about zero and spans roughly $\pm 30,000$ — about seven times the empirical range — confirming the weakly informative character of the specification. A portion of prior mass falls below zero, which is physically inadmissible but acceptable as an artefact of unconstrained Gaussian priors on $\boldsymbol{\beta}$; this region carries negligible posterior probability once the data are conditioned on.

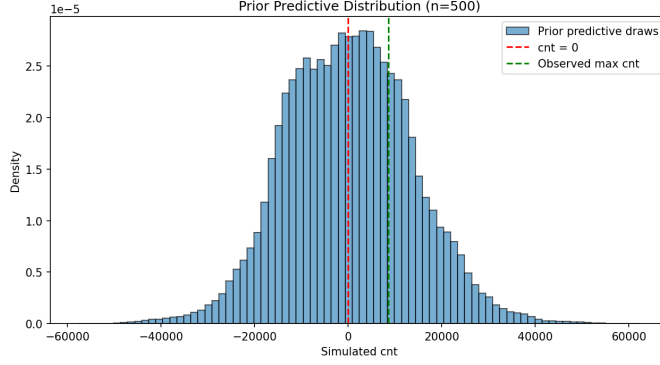


Figure 1: Prior predictive distribution of y from 500 joint-prior draws. Dashed lines mark $y = 0$ and the maximum observed value 8555.

0.2 Posterior Inference

Combining the likelihood (1) with the priors (2) yields the joint posterior $p(\boldsymbol{\beta}, \sigma \mid \mathbf{y}, \mathbf{X})$ over $(\beta_0, \beta_1, \beta_2, \sigma)$. This posterior admits no closed form: the joint presence of three coefficients and an unknown residual scale precludes the Normal–Normal conjugacy available in the single-coefficient known-variance case. We therefore approximate it by Markov chain Monte Carlo (MCMC).

MCMC configuration (Core Task 1, §3.3) Sampling was performed with the No-U-Turn Sampler (NUTS), a self-tuning variant of Hamiltonian Monte Carlo, as implemented in PyMC 5. We ran $C = 4$ independent chains with $W = 1,000$ warm-up iterations and $T = 2,000$ retained iterations per chain, producing $S = CT = 8,000$ posterior samples. The warm-up phase serves a dual role: NUTS adapts its step size and diagonal mass matrix during this phase, and the corresponding draws are simultaneously discarded as burn-in, so no further explicit burn-in is removed. The target acceptance probability was set to $\alpha = 0.95$ (above the default 0.80) to suppress divergent transitions; none were reported. The random seed was fixed at 42, matching the data-subsampling seed of Module A, so that the entire pipeline is reproducible from raw data to posterior summaries.

Convergence diagnostics Two standard diagnostics confirm that the four chains have converged to a common posterior distribution. First, the potential scale reduction factor satisfies $\hat{R} = 1.00$ for every parameter, well below the conventional warning threshold of 1.01. Second, the bulk effective sample size satisfies $N_{\text{eff}}^{\text{bulk}} \in [3,775, 5,371]$ across the four parameters, exceeding the rule-of-thumb minimum of 400 recommended for stable estimation of central tendency. A more detailed treatment of trace plots and autocorrelation is deferred to Section ; the present subsection therefore proceeds directly to posterior inference.

Posterior summaries Table 2 reports, for each parameter $\theta \in \{\beta_0, \beta_1, \beta_2, \sigma\}$, the posterior mean $\hat{\theta} = \mathbb{E}[\theta \mid \mathbf{y}]$, the posterior standard deviation, and the 95% highest-density credible interval (HDI). The HDI is preferred over an equal-tailed interval because it always contains the most plausible parameter values and remains meaningful for asymmetric posteriors. Marginal posterior densities of all four parameters are shown in Figure 2; each is approximately symmetric and unimodal.

Table 2: Posterior summaries based on $S = 8,000$ NUTS samples (4 chains $\times$ 2,000 retained draws). All $\hat{R} = 1.00$ and bulk ESS exceeds 3,500 for every parameter.

Parameter	Mean	SD	95% HDI low	95% HDI up	ESS bulk	$\hat{R}$
β_0 (intercept)	2541	382	1799	3324	3,775	1.00
β_1 (x_{i1} , temp.)	6784	436	5915	7632	5,371	1.00
β_2 (x_{i2} , hum.)	−2350	555	−3457	−1276	4,181	1.00
σ	1500	56	1393	1610	5,352	1.00

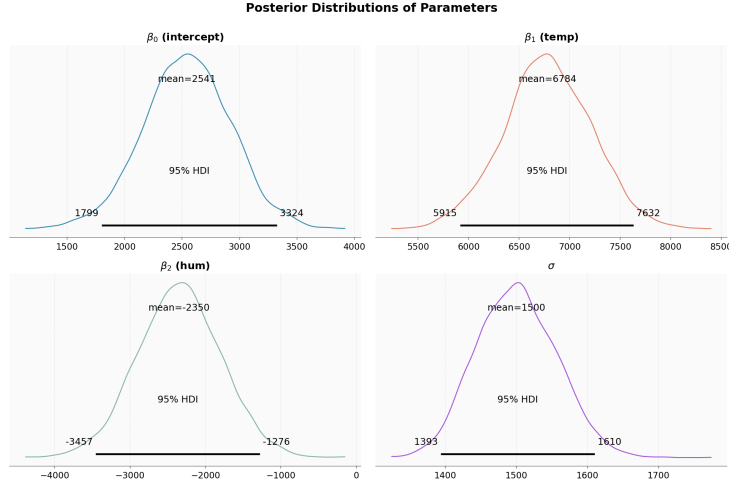


Figure 2: Marginal posterior densities of $\beta_0, \beta_1, \beta_2, \sigma$; the bar beneath each curve marks the 95% HDI.

Substantive interpretation (Core Task 1, §3.4) Coefficient interpretations refer to the normalised covariate scale: a unit change in x_{i1} corresponds to the full empirical temperature swing (-8 to 39°C , a 47°C range), and a unit change in x_{i2} corresponds to the full humidity swing (from arid to saturated air).

Temperature (β_1). The posterior mean is $\hat{\beta}_1 = 6784$ with 95% HDI (5915, 7632). The lower limit exceeds zero by more than 13 posterior standard deviations, so the posterior probability $\Pr(\beta_1 > 0 \mid \mathbf{y})$ is effectively unity. A unit increase in normalised temperature is associated, on average, with approximately 6,800 additional daily rentals, identifying temperature as the dominant explanatory mechanism in the model.

Humidity (β_2). The posterior mean is $\hat{\beta}_2 = -2350$ with 95% HDI ($-3457, -1276$). The exclusion of zero from this interval supports the conclusion that humidity meaningfully suppresses bike-sharing activity, in agreement with the negative slope visible in the exploratory analysis of Section . The wider interval relative to that of β_1 reflects both the weaker correlation in the data and the smaller absolute effect.

Intercept (β_0). The posterior mean is $\hat{\beta}_0 = 2541$ with 95% HDI (1799, 3324). Under the parametrisation in (1), β_0 represents the expected daily count when both predictors are simultaneously zero — a corner of the predictor space not realised in the data, retained for completeness rather than substantive interpretation.

Residual scale (σ). The posterior mean is $\hat{\sigma} = 1500$ with 95% HDI (1393, 1610). The relative scale $\hat{\sigma}/\bar{y} \approx 0.34$ indicates that temperature and humidity together leave roughly one third of the daily standard deviation of y unexplained, consistent with the modest coefficient of determination $R^2 = 0.405$ obtained from the OLS baseline. This residual variation is attributable to calendar effects, holidays, and other meteorological covariates that are deliberately omitted in order to retain the two-predictor specification dictated by the project scope.